

Social Interactions

Unit 4 — Social Interactions & Game Theory

- › Games, best responses & the Nash equilibrium
- › Dominant strategies & the prisoner's dilemma
- › Resolving social dilemmas: altruism, repetition, rules
- › Sequential games & coordination

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 4 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. Every payoff matrix is redrawn in the KU Leuven bi-matrix style (row player's payoff bottom-left, column player's top-right), Nash cells outlined in green, so you can read the strategic logic at a glance.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

Previous units modelled choices that **did not depend on other people's decisions**. Adam Smith argued that self-interested individuals, guided by the "**invisible hand**", can produce outcomes good for society (entrepreneurship, innovation). But self-interest can also be **harmful**. This unit uses **game theory** to model interactions where my payoff depends on your action, and asks: why do these problems arise, and what can we do about them?

DEFINITION — SOCIAL DILEMMA

A **social dilemma** is a situation in which actions taken **independently by self-interested individuals** result in a **socially sub-optimal outcome** — e.g. traffic jams, climate change, COVID-19. Two classic forms:

- **Tragedy of the commons** — common resources are **over-exploited** (Lloyd 1833, overgrazing; Hardin 1968). E.g. fishing, climate, a shared kitchen.
- **Free riding** — benefiting from others' contributions to a cooperative project **without contributing yourself**.

The unit asks how **altruism**, **repetition**, and **government policy** can resolve them.

Outline

1. **A.** Introduction — self-interest, social dilemmas
2. **B.** Game theory: key concepts — games, Nash equilibrium, dominant strategies, prisoner's dilemma
3. **C.** Evaluating outcomes — the Pareto criterion
4. **D.** Resolving social dilemmas — social preferences, repeated games, changing the rules
5. **E.** Coordination — multiple equilibria

SECTION B

Game Theory: Key Concepts

DEFINITIONS

- **Social interaction** — a situation involving more than one party, where one's actions affect **both their own and others'** outcomes.
- **Strategic interaction** — a social interaction where people are **aware** of how their actions affect others.
- **Strategy** — an action a player can take in a social interaction.

Games are studied with **game theory** (used in social science *and* biology).

DEFINITION — A GAME

A **game** describes a social interaction through five elements:

1. **Players** — who is involved.
2. **Feasible strategies** — the actions each player can take.
3. **Order of play** — what each player knows when deciding.
4. **Information** — what each knows when choosing an action.
5. **Payoffs** — outcomes for every combination of actions.

Worked example — crop choice

Two farmers, **Anil** and **Bala**, each choose which crop to specialise in — **Rice** or **Cassava**. They interact only once (a **one-shot game**) and choose **simultaneously** (neither knows the other's choice). Payoffs depend on market prices and land quality.

		Bala	
		Rice	Cassava
Anil	Rice	4, 4	6, 3
	Cassava	6, 6	5, 2

Payoffs: **bottom-left** = Anil (row), **top-right** = Bala (column).

Best response & Nash equilibrium

DEFINITIONS

- **Best response** — the strategy that yields the highest payoff, **given the other player's strategy**.
- **Nash equilibrium** — a set of strategies (one per player) such that **each player's strategy is a best response to everyone else's**. No player can gain by **deviating unilaterally**. (After John Nash, 1928–2015, Nobel 1994 — see *A Beautiful Mind*.)

Finding the Nash equilibrium of the crop game by checking best responses:

- If Bala plays Rice, Anil's best response is **Cassava** ($6 > 4$).
- If Bala plays Cassava, Anil's best response is **Rice** ($6 > 5$).
- If Anil plays Rice, Bala's best response is **Rice** ($4 > 3$).
- If Anil plays Cassava, Bala's best response is **Rice** ($6 > 5$).

		Bala	
		Rice	Cassava
Anil	Rice	4 / 4	3 / 6
	Cassava	6 / 6	2 / 5

The unique Nash equilibrium (green): **Anil grows Cassava, Bala grows Rice** → payoffs (6, 6). Here it is also Pareto-efficient — but that need not be true in general.

LISTEN FOR IN THE LECTURE

Reading a bi-matrix: to find **Anil's** best response, fix a Bala **column** and compare the **bottom-left** numbers; to find **Bala's**, fix an Anil **row** and compare the **top-right** numbers. A cell is a Nash equilibrium when it is best for *both* at once. **A game can have more than one Nash equilibrium** — or none in pure strategies.

Finding Nash Equilibria

A 3×3 example

Player 1 chooses a row (T/M/B); Player 2 a column (L/C/R). Find every best response:

		Player 2		
		L	C	R
Player 1	T	1 2	2 0	3 0
	M	1 1	1 1	0 1
	B	1 0	0 2	2 2

Bottom-left = Player 1, top-right = Player 2. Unique Nash equilibrium: **(B, R)** → (2, 2).

- If Player 2 plays **L**, Player 1's best is **T** ($2 > 1 > 0$).
- If Player 2 plays **C**, Player 1's best is **B** ($2 > 1 > 0$).
- If Player 2 plays **R**, Player 1's best is **B** ($2 > 1 > 0$).
- Likewise for Player 2. **Only in cell (B, R)** are both strategies best responses to each other — no player can improve by unilaterally changing. That is the **Nash equilibrium**.

Dominant strategies & the prisoner's dilemma

DEFINITIONS

- **Dominant strategy** — a strategy that is the best response **regardless of what the other player does** (does not always exist).
- **Dominant strategy equilibrium** — a Nash equilibrium in which **every** player plays a dominant strategy.

Back to Anil and Bala, now deciding how to deal with pest insects: **Toxic Tide** (kills every insect but pollutes) vs **Integrated Pest Control / IPC** (less effective, no pollution).

		Bala	
		IPC	Toxic Tide
Anil	IPC	3 / 3	1 / 4
	Toxic Tide	4 / 1	2 / 2

No matter what the other does, **Toxic Tide** beats IPC for each player (a dominant strategy) → equilibrium (Toxic, Toxic) = (2, 2).

DEFINITION — PRISONER'S DILEMMA

A game with a **dominant-strategy equilibrium** in which playing the dominant strategy yields **lower individual AND total payoffs** than another (cooperative) outcome. Here both would be better off at (IPC, IPC) = (3, 3), but self-interest drives them to (Toxic, Toxic) = (2, 2). (Originally told as two burglars caught by police.)

CHECK YOURSELF · EXAM MCQ

Which is correct?

- (a) "An equilibrium in dominant strategies is always a Nash equilibrium." — **True**. If each plays a best response regardless of the other, it is certainly a best response to what the other actually does.
- (b) "The prisoner's dilemma outcome is never a Nash equilibrium, as all players can improve by changing strategies." — **False**. It is a Nash equilibrium; no one can improve by changing *unilaterally* (both must change together to reach the better outcome).
- (c) "A strategy combination that maximises the sum of payoffs is always a Nash equilibrium." — **False**. The efficient (IPC, IPC) maximises the total but is *not* a Nash equilibrium.

Answer: (a).

SECTION C

Evaluating Outcomes — the Pareto Criterion

DEFINITIONS — PARETO

- Allocation A **Pareto-dominates** B if **at least one party is better off under A and nobody is worse off**.
- An allocation is **Pareto-efficient** if **nobody can be made better off without making somebody worse off**.

Often **many** outcomes are Pareto-efficient — e.g. every way of splitting €1000 between two people is efficient as long as no money is left on the table.

Plot the four pest-control outcomes in **payoff space** (Anil's payoff on the x-axis, Bala's on the y-axis):

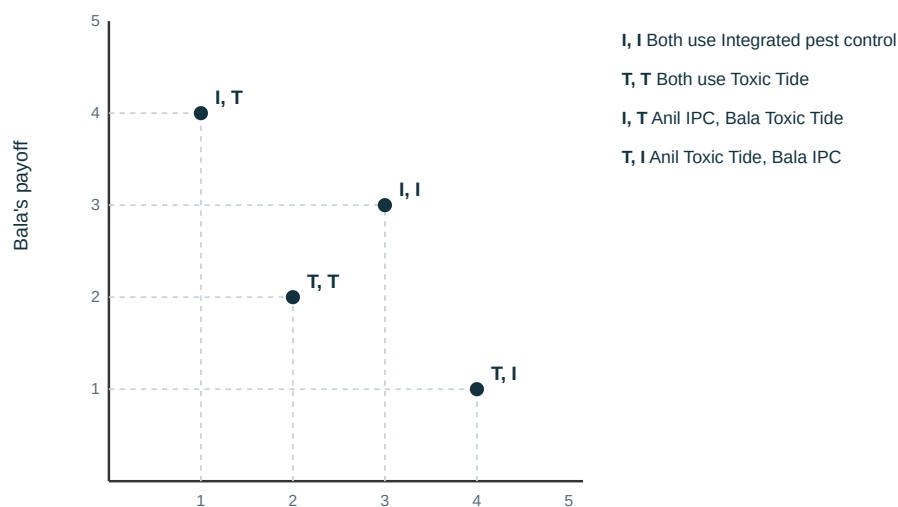


Figure 1. The four outcomes of the pest game. $(I, I) = (3, 3)$ Pareto-dominates $(T, T) = (2, 2)$ — both do better. But (I, I) does **not** dominate (I, T) or (T, I) , where one player does better and the other worse. (I, T) , (I, I) and (T, I) are all Pareto-efficient.

Caveats of the Pareto criterion

TWO IMPORTANT LIMITS

- **It often can't choose.** With many efficient allocations, the Pareto criterion is silent about which to pick. It may not even rank an efficient against an inefficient one — e.g. (T, I) is efficient but does **not** Pareto-dominate the inefficient (T, T).
- **It says nothing about fairness.** Many **unfair** allocations are Pareto-efficient — e.g. keeping €99.99 of a found €100 and giving your friend 1 cent is efficient (no money wasted) yet hardly fair.

LISTEN FOR IN THE LECTURE

Pareto efficiency is a **low bar**: "no waste", not "good" or "fair". The Nash equilibrium of a prisoner's dilemma is **Pareto-inefficient** (both could do better) — which is exactly why we look for ways to escape it in Section D.

SECTION D

Resolving Social Dilemmas

The prisoner's-dilemma outcome arose for three reasons. Each suggests a route out:

THREE ROUTES OUT OF A SOCIAL DILEMMA

1. Players **only cared about their own payoffs** → introduce **social preferences / altruism**.
2. Nobody could make players **pay for the consequences** of their actions on others → introduce **repeated games, social norms, peer punishment**.
3. Players **could not coordinate** beforehand → **change the rules of the game** (institutions and policies).

D.i — Social preferences: altruism

Social/prisoner's dilemmas arise when players care **only** about their own payoffs. But in experiments many players show **altruism** — they choose the dominated (cooperative) strategy. If altruism is **strong enough**, the socially desirable outcome can be reached (in *Golden Balls*, (split, split) is common; likewise in public-good games).

How altruistic are we? The public-good game

PUBLIC-GOOD GAME (PRISONER'S DILEMMA FOR N PLAYERS)

Each of **n** players gets **T** tokens and chooses how many to keep (z_i) vs contribute to a common pool ($g_i = T - z_i$). The payoff is:

$$\text{payoff}_i = z_i + \frac{\beta}{n}(g_1 + g_2 + \dots + g_n)$$

- If $1 > \beta/n$, a purely self-interested player **keeps all tokens** (private return on a contributed token, β/n , is below 1).
- If $\beta > 1$, it is **socially optimal to contribute everything** (each token multiplied by $\beta > 1$ across the group).
- For $1 > \beta/n$ and $\beta > 1$ together, the game is a **social/prisoner's dilemma**.

Experimental finding: in public-good games worldwide ($\beta = 1.6$, $n = 4$, 20 tokens each), people start by contributing a **lot** (≈ 8 – 14 of 20), far above the self-interested prediction of zero — clear evidence of **altruism/cooperation**. But without punishment, contributions **decline period by period** as cooperators get discouraged by free-riders.

Modelling altruistic preferences

Represent preferences with **indifference curves** over the two players' payoffs and combine with the feasible set (as in Unit 3). Example: Zoë wins **£200** in a lottery — will she keep it all, or share with her flatmate Yvonne?

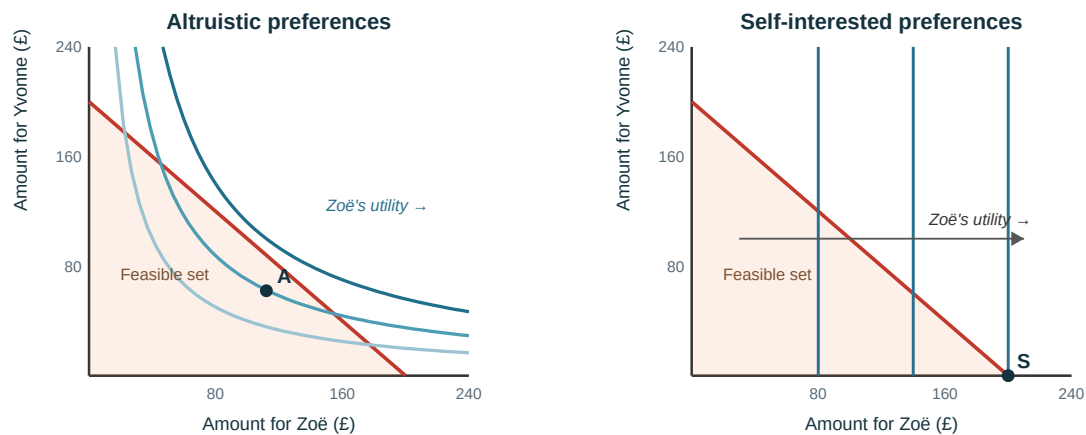


Figure 2. Altruistic Zoë (left) has **convex indifference curves** over money-for-Zoë and money-for-Yvonne, so her optimum **A** shares some money (interior point on the £200 budget line). **Self-interested Zoë** (right) has **vertical** indifference curves — she only values her own money — so her optimum **S** keeps all £200 (corner).

Repeated Games & Punishment

So far, **one-shot** games. Better outcomes can arise in **repeated interactions** through **social norms**, **reciprocity**, and **peer punishment**: behaving selfishly today has **consequences in future periods**, so it may **no longer be a dominant strategy**.

Worked example — the cartel's dilemma (two petrol stations)

Jupiter and Saturn each choose to **Maintain** a high price or **Cut** it. In a **one-shot** game, cutting is a dominant strategy for both → both cut (the **cartel's dilemma**).

		Saturn	
		Maintain	Cut
Jupiter	Maintain	1000 / 1000	700 / 1300
	Cut	1300 / 700	800 / 800

Profits (bottom-left = Jupiter, top-right = Saturn). One-shot Nash: **(Cut, Cut) = (800, 800)** — worse for both than (Maintain, Maintain) = (1000, 1000).

TRIGGER STRATEGY IN THE REPEATED GAME

When the game repeats, a firm can follow a "**trigger**" strategy: start with the **high** price, and keep it as long as the rival does too; if the rival **cuts**, retaliate with a low price for a while, then return to the high price. This creates a **trade-off**: high profit **today** from cutting vs high profits in the **future** from maintaining. Cooperation (Maintain, Maintain) can be an **equilibrium** if players value future money highly enough — because **punishment becomes possible**.

Punishment sustains cooperation — the evidence

In the public-good game, contributions normally **decline** over the rounds. But when players are allowed to **punish** free-riders (at a cost to themselves), contributions become **much higher and stay high** — peer punishment sustains cooperation. The willingness to punish, even at personal cost, is itself a social preference.

LISTEN FOR IN THE LECTURE

The key insight: repetition turns a one-shot prisoner's dilemma into a game where **defection can be punished later**, so the dominant strategy to defect disappears. This is how **social norms, reputation and reciprocity** support cooperation in real life without any government.

Rules of the Game — Sequential Games

The third route: **change the rules**. One important rule is the **order of play**. When players move in **sequence** (observing earlier choices), the game is drawn in **extensive form** (a **game tree**).

DEFINITIONS

- **Extensive form (game tree)** — **nodes** are points where a player picks a move; **branches** are the possible choices.
- **Backward induction** — solve by first finding the optimal choice at the **final** nodes, then reasoning **backward** to the first node.

Worked example — two TV stations

Channels **Z** and **Delta** choose a time slot (7:30 or 8:00 pm) for the evening news. **Z chooses first**, and Delta observes Z's choice before deciding.

Green = backward-induction path → outcome (Z:4, D:3)

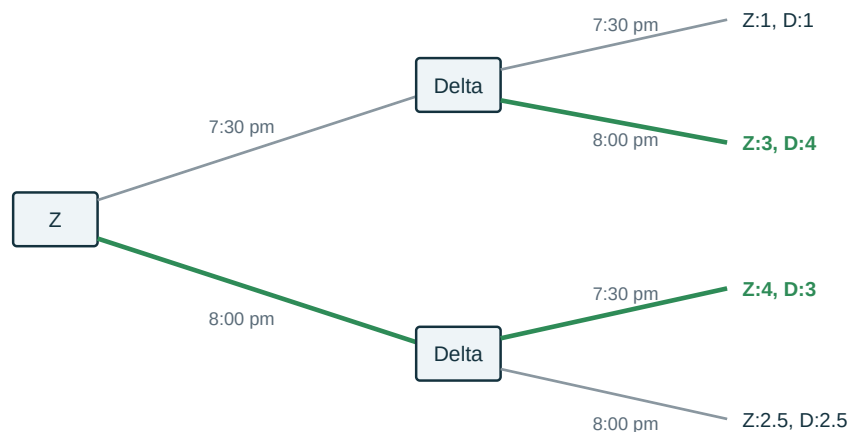


Figure 3. Solve by **backward induction**. If Z picks 7:30, Delta prefers 8:00 ($D:4 > D:1$) → (Z:3, D:4). If Z picks 8:00, Delta prefers 7:30 ($D:3 > D:2.5$) → (Z:4, D:3). Anticipating this, Z compares 3 vs 4 and picks **8:00**. Outcome: **(Z:4, D:3)**.

The ultimatum game

DEFINITION — ULTIMATUM GAME

A **sequential game** for dividing an economic rent (e.g. a cash prize). The **Proposer** offers a split; the **Responder** observes it and **accepts** (both get the proposed amounts) or **rejects** (both get **nothing**).

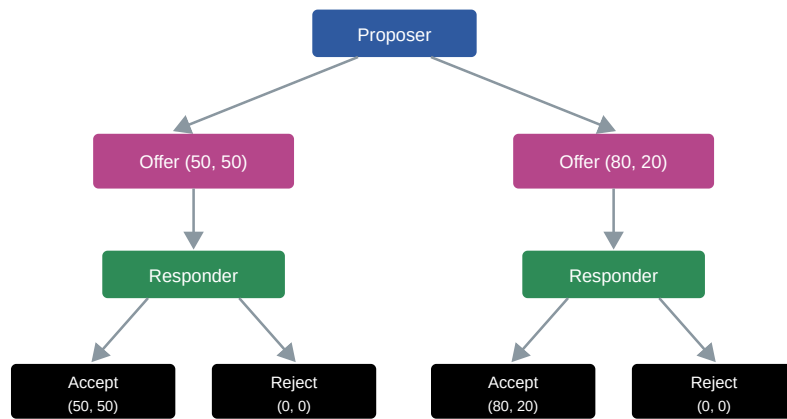


Figure 4. The ultimatum game in extensive form. If players cared **only about their own payoffs**, the Nash equilibrium (by backward induction) is that the Proposer offers **almost nothing** and the Responder **accepts** (since anything > 0 beats 0). In reality, offers are often near **50–50**.

WHY REAL OFFERS ARE GENEROUS

The Proposer's offer may be motivated by **altruism, fairness (50–50), inequality aversion, social norms, or reciprocity**. Responders routinely **reject** low offers — punishing unfairness at a cost to themselves. **Evidence (Kenyan farmers vs US students):** offers are consistent with social preferences *but also* with expected-payoff maximisation (proposers offer more where low offers are more likely to be rejected). And **the rules matter:** introducing **competition between responders** (two responders vying to accept) pushes outcomes back toward the **self-interested** prediction.

SECTION E

Coordination — Multiple Equilibria

Not every social dilemma is a prisoner's dilemma. Some games have **several** Nash equilibria, and the problem is **coordinating** on one of them.

Driving on the left or the right

Two drivers approach each other; each chooses to drive on the Left or the Right. If they choose the same side, all is well (payoff 0); if they mismatch, they crash (−10 each).

		Driver 2	
		Left	Right
Driver 1	Left	0, 0	−10, −10
	Right	−10, −10	0, 0

Two Nash equilibria: (Left, Left) and (Right, Right). Coordination is possible — and highly important.

COORDINATION GAMES & INSTITUTIONS

With **multiple equilibria**, no player has an incentive to deviate from either, so the outcome is **indeterminate** without coordination. **Institutions and conventions** select the equilibrium: a traffic law (drive on the right), or a coordinated switch. Sweden's "**Dagen H**" (**3 Sept 1967**) — the whole country switched from left to right overnight — is a real example of institutions moving everyone from one equilibrium to another.

Climate change as a game

For purely self-interested players, "**business as usual**" (**BAU**) is a dominant strategy (a prisoner's dilemma), and an emissions treaty would deliver the socially optimal outcome. But if players have **inequality aversion and reciprocity**, the game can become a **coordination game** with **two Nash equilibria**, one of which is the **social optimum** (both restrict).

		US	
		Restrict	BAU
China	Restrict	BEST BEST	GOOD WORST
	BAU	WORST GOOD	BAD BAD

With social preferences, (Restrict, Restrict) and (BAU, BAU) are both Nash equilibria — so coordination (a treaty) can move both countries to the **BEST** outcome.

LISTEN FOR IN THE LECTURE

The type of game depends on **preferences**: pure self-interest → prisoner's dilemma (one bad equilibrium); social preferences → coordination game (a good equilibrium becomes available). The lecturer's point: **changing preferences or rules changes the game itself**, and institutions exist to steer us toward the good equilibrium (cf. the Dvorak keyboard, another coordination trap).

Summary, Formulas & Glossary

The unit in one paragraph

Social interactions are modelled as **games** (players, strategies, order, information, payoffs). Players choose **best responses**, and a **Nash equilibrium** is a profile where everyone is best-responding — nobody gains by deviating alone. A **dominant strategy** is best whatever others do; when a dominant-strategy equilibrium leaves everyone **worse off** than a cooperative outcome, it is a **prisoner's dilemma** — a Pareto-inefficient Nash equilibrium. The **Pareto criterion** only flags "no waste" (not fairness) and often can't rank outcomes. Social dilemmas can be escaped through **social preferences/altruism**, **repetition with punishment** (trigger strategies, reputation), or **changing the rules** (sequential order, institutions). Some games instead have **multiple equilibria**, where the challenge is **coordination**, and institutions select the good equilibrium.

Concept sheet

Concept	Meaning
Game	players, strategies, order, information, payoffs
Best response	highest-payoff strategy given others' strategies
Nash equilibrium	each plays a best response; no profitable unilateral deviation
Dominant strategy	best response regardless of others' choices
Prisoner's dilemma	dominant-strategy eq. with lower individual & total payoff than cooperation
Pareto dominance / efficiency	A better for someone, worse for none / no waste
Public-good game	$\text{payoff} = z_i + (\beta/n) \cdot \Sigma g$; dilemma when $1 > \beta/n$ and $\beta > 1$
Repeated game / trigger	future punishment can sustain cooperation
Backward induction	solve sequential game from the final nodes back
Coordination game	multiple Nash equilibria; institutions select one

Glossary

Term	Meaning
Social / strategic interaction	Actions affect others / players are aware they do.
Social dilemma	Independent self-interest → socially sub-optimal outcome.
Tragedy of the commons / free riding	Over-exploiting shared resources / benefiting without contributing.
Best response / Nash equilibrium	Optimal reply / mutual best responses.
Dominant strategy (equilibrium)	Best whatever others do (all players do so).
Prisoner's dilemma	Dominant-strategy eq. worse for all than cooperating.
Altruism / social preferences	Caring about others' payoffs, fairness, reciprocity.
Repeated game / peer punishment	Interaction over many periods / costly punishment of defectors.
Extensive form / backward induction	Game tree / solving from the end.
Ultimatum game	Proposer splits a rent; responder accepts or rejects.
Coordination game	Multiple equilibria; the problem is agreeing which.

Exam-style questions to practise

1. Define a Nash equilibrium. In the crop game, verify that (Cassava, Rice) is Nash by checking both players' best responses.
2. Why is the prisoner's-dilemma outcome a Nash equilibrium *and* Pareto-inefficient? Reconcile these with a 2×2 matrix.
3. In a public-good game with $\beta = 1.6$ and $n = 4$, show that a self-interested player contributes zero but full contribution is socially optimal.
4. Explain how a trigger strategy can sustain the (Maintain, Maintain) outcome in the petrol-station game. What condition on the value of future profits is needed?
5. Solve the two-TV-station game by backward induction. Why does moving first benefit Channel Z?
6. Give the two Nash equilibria of the driving game. Why can't the Pareto criterion pick between them, and what selects the outcome in practice?

the crop, pest-control, cartel, climate and driving games reproduce the slides' payoffs, and the sequential and ultimatum trees the slides' extensive forms.